

Ashutosh Swain

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SUMMARY

Computer Science undergraduate focused on backend and distributed systems. I learn by building from scratch — coordination services, BitTorrent clients, schema design tools — and shipping things that solve real problems. Currently exploring infrastructure tooling and MCP servers. [See all repositories](#).

EXPERIENCE

Commenda — Software Engineering Intern

[commenda.io](#)

Working on reliability and backend systems. Contributing across the stack with a focus on resilient services, observability, and developer ergonomics.

PROJECTS

casper — MCP server for Terraform infrastructure

[Demo](#) | [Repo](#)

An MCP server that gives AI agents a live, queryable view of Terraform infrastructure. Supports graph queries, drift detection against AWS, policy gates, and blast-radius simulation before `terraform apply`. Built in Go with a graph-backed query layer over parsed HCL and remote state.

skyforge — Collaborative DB schema design tool

[Demo](#) | [Repo](#)

Full-stack platform to design, visualize, and share database schemas with real-time collaboration over ER diagrams. Go backend for schema parsing and code generation across multiple SQL dialects, Next.js frontend with a custom canvas for diagram editing.

maxim — TUI client for local databases

[Repo](#)

A terminal UI for Postgres built on Bubble Tea. Schema-aware SQL editor with autocomplete, secure credential handling, and integrated Docker management for one-click local database spin-up and teardown. Achieved 138 stars on github on this project

torrent — BitTorrent client in Go

[Repo](#)

End-to-end BitTorrent client implementing bencode parsing, the peer wire protocol, piece selection, and swarm orchestration. Read the spec, shipped the bytes.

EDUCATION

2024 - 2028 B.Tech (Information Technology) at **Vellore Institute of Technology**

WRITING

Understanding Amazon's DynamoDB Architecture. A breakdown of DynamoDB's distributed design — partitioning, replication, and the trade-offs behind its consistency model. [Link](#)

The BitTorrent Protocol. Written alongside building a BitTorrent client in Go — peer-to-peer networking, piece exchange, and protocol-level design. [Link](#)

Understanding Google's Borg. How Borg manages cluster-scale scheduling, resource allocation, and fault tolerance — the ideas that inspired Kubernetes. [Link](#)

SKILLS

Languages	Go, TypeScript, JavaScript, Rust, SQL
Backend & Systems	Node.js, Express, WebRTC, Redis, gRPC, REST APIs, Bubble Tea (TUI)
Frontend	Next.js, React, Tailwind CSS
Infrastructure	Terraform, AWS, Docker, MCP, Postgres, Cloudflare R2, Backblaze B2